

ARUN S

Full-Stack Developer • MCA Student • ML Enthusiast
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[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

CAREER OBJECTIVE

Seeking a Software Developer or Full-Stack Engineer role to apply expertise in React.js, Next.js, Node.js, PostgreSQL, and Python-based Machine Learning — building scalable, high-performance digital products while growing into a senior engineering position.

PROFESSIONAL SUMMARY

Full-Stack Developer and MCA student at Sathyabama Institute of Science and Technology with 3+ production web applications and a Python Machine Learning internship. Experienced in React.js, Next.js, Node.js, Express.js, PostgreSQL, and MongoDB. Key results: 40% faster recruiter screening, sub-100ms query response times, and 30% faster UI development. Proficient in REST API design, JWT Authentication, clean code, and Agile development.

TECHNICAL SKILLS

Frontend : React.js (Hooks, JSX), Next.js, HTML5, CSS3, Tailwind CSS, Framer Motion, JavaScript (ES6+)
Backend : Node.js, Express.js, REST API Design, JWT Authentication, Middleware Architecture
Databases : PostgreSQL, MongoDB, MySQL, Mongoose ODM, SQL
ML / AI : Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn — Model Training, Data Analysis
Programming : Python, JavaScript, Java, C
Tools & DevOps : Git, GitHub, Vercel, Netlify, Streamlit, VS Code, Postman, npm, yarn

PROJECTS

LEVEL UP — AI-Powered Career & Recruitment Platform

Tech Stack: Next.js | Node.js | Express.js | PostgreSQL | JWT Auth

- Gamified recruitment platform for a 5-member team — replaced resume-only hiring with skill-progression profiles, reducing recruiter screening time by 40%.
- Built role-based dashboards (Candidate, Recruiter, Admin) with AI-enhanced profile analysis supporting 3 user roles in a single Next.js codebase.
- Designed PostgreSQL relational models for skill rankings, user roles, and recruiter workflows with optimised multi-table queries.
- Lazy loading and reusable component architecture across 15+ pages improved UI development speed by ~30%.

FurnitureMart — Full-Stack E-Commerce Web Application

Tech Stack: Node.js | Express.js | MongoDB | HTML/CSS/JS | JWT Auth

- Built a full-stack e-commerce platform with 100+ SKUs, category filtering, cart management, and secure checkout — zero manual order errors.
- Implemented JWT authentication with role-based route protection; 2 user roles and 8+ secured REST API endpoints.
- Admin CRUD panel for products, users, and orders reduced update time by 50%; MongoDB indexing achieved sub-100ms query response times.

Smart To-Do App — Task Management Application

Tech Stack: HTML | CSS | JavaScript | Local Storage API

- Responsive task management app with 3-level priority filters and Local Storage persistence — 100% data retention across sessions with no backend.
- Sub-50ms interaction response times with Lighthouse performance score of 95+.

RF Signal Prediction System — Machine Learning Project

Tech Stack: Python | Scikit-learn | Pandas | NumPy | Matplotlib

- Trained Linear Regression and Random Forest models on 1,000+ records — Random Forest achieved 89% R² score with 22% RMSE reduction via feature engineering and 5-fold CV.
- Deployed on Streamlit for real-time RF signal strength predictions from user-supplied inputs.

INTERNSHIP EXPERIENCE

Machine Learning with Python Intern

June 2024

Retech Solutions Pvt. Ltd., Tambaram West, Chennai

- Built and deployed 3+ supervised ML models (classification and regression) using Python and Scikit-learn — achieving up to 89% accuracy on held-out test sets.
- Automated data cleaning pipelines (imputation, encoding, normalisation) using Pandas and NumPy — reduced preprocessing time by 35% across 5,000+ row datasets.
- Improved model F1-score by 18% through hyperparameter tuning and k-fold cross-validation; documented and presented reports to senior engineers.
- Visualised feature importance and model outputs using Matplotlib and Seaborn for non-technical stakeholders.

EDUCATION

Master of Computer Applications (MCA)	2025 – Present
Sathyabama Institute of Science and Technology, Chennai	Pursuing
Bachelor of Computer Applications (BCA)	2022 – 2025
Guru Nanak College, Chennai	81.10%
HSC – Class XII	2021 – 2022
SSV HR Sec School, Mathapattanam	66.5%
SSLC – Class X	2019 – 2020
SSV HR Sec School, Mathapattanam	61.2%

CERTIFICATIONS

- MongoDB Certified Developer — Proficiency in NoSQL schema design, aggregation pipelines, indexing, and performance tuning.
- Cybersecurity Essentials — Cisco Networking Academy: network security, threat mitigation, cryptography, and best practices.

KEY STRENGTHS

- Problem-Solving and Analytical Thinking — Consistently debugs complex issues across frontend, backend, and ML pipelines.
- Agile Self-Learner — Rapidly adopts new frameworks and technologies through hands-on project-based learning.
- Team Collaboration — Experienced in code reviews, peer feedback, and collaborative academic and internship projects.
- Attention to Detail — Strong focus on clean code, UI precision, and thorough testing before every deployment.